

Supplementary Information for

Recipe determinants of quantization fragility under image corruption:

paired evidence and a regression-head localization in object detectors

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S1 Purpose and scope

This supplement documents the frozen protocol details, estimands, corruption parameters, calibration schedules, and artifact contract supporting the main article. All reported values are generated deterministically from retained ledgers; nothing here adds a claim beyond the main text.

S2 Estimands

For dataset d , model m , corruption condition c , and a pair of treatments (A, B) sharing the same evaluator:

$$G_0 = \text{AP}_A(\text{J95}) - \text{AP}_B(\text{J95}), \quad (\text{S1})$$

$$G_c = \overline{\text{AP}_A(c) - \text{AP}_B(c)} \quad (\text{equal weight over the 12 cells}), \quad (\text{S2})$$

$$\Delta E = G_c - G_0. \quad (\text{S3})$$

J95 denotes the codec-matched clean control: the J95 clean control and every corrupted input are terminally encoded as JPEG quality 95 with chroma subsampling disabled, so the clean control and the corrupted inputs differ only in the named corruption transform, not in container encoding; the separate original-clean cell keeps the dataset-native bytes and is used only to check control fidelity. $\Delta E > 0$ means the A-minus-B gap widens under corruption; $\Delta E < 0$ means it narrows. A narrowing gap near a shared accuracy floor is not read as robustness: absolute corrupted AP is reported alongside every interaction (main text, §4.1–§4.3).

S3 Frozen condition schedule

Each dataset×model block evaluates 14 conditions on identical image ordering for quantized arms (13 for FP32 and INT8-legacy arms that predate codec control):

- clean (original bytes) and J95 (JPEG-95 clean control);
- Gaussian noise: $\sigma \in \{8, 20, 40\}$;
- motion blur: kernel lengths $\{5, 13, 23\}$;
- synthetic fog: (strength, radius) $\in \{(0.18, 6), (0.38, 12), (0.62, 20)\}$;
- JPEG compression: qualities $\{70, 40, 15\}$.

Corruption parameters are frozen in versioned configuration `mn.paired_protocol_v1.20260924`; the same transform code and the same JPEG-95 terminal encoding produce both the evaluation caches and the corruption-aware calibration images.

S4 Treatments and arm registry

Table S1: RetinaNet arm registry. “Calibration” lists the input distribution seen by entropy calibration; “sites” lists the quantized region.

Arm	Precision	Quantized sites	Calibration input
FP32	FP32	none	—
FP8-matched	FP8	full graph	clean, matched preprocessing
INT8-legacy	INT8	full graph	mismatched preprocessing (diagnostic)
INT8-matched	INT8	full graph	clean, 512 images
INT8-q95calib	INT8	full graph	J95 re-encodes, 512 images
INT8-selective	INT8	all except <code>/head/regression.head/</code>	clean, 512 images
INT8-sel-q95calib	INT8	selective	J95 re-encodes, 512 images
INT8-corruptcalib	INT8	full graph	fold-1 corruptions (noise, JPEG), 512 images
INT8-sel-corruptcalib	INT8	selective	in-family corruptions, 512 images
INT8-cc2calib	INT8	full graph	fold-2 corruptions (fog, motion blur), 512 images
INT8-sel-cc2calib	INT8	selective	fold-2 corruptions, 512 images

All INT8/FP8 engines are built by TensorRT 11.1.0.106 from ONNX graphs quantized by NVIDIA ModelOpt 0.45.0 (entropy calibration, 512-image budget), with TF32 disabled and a 4,096 MiB builder workspace. Calibration and evaluation images are disjoint. Checkpoints are one per dataset×model cell: YOLO11n/m/x are fine-tuned from Ultralytics pretrained weights on each dataset’s locked confirmatory training split (640-pixel input; per-run training manifests are hash-bound), and the torchvision RetinaNet-R50-FPN-v2 model is fine-tuned per dataset for 100 epochs at 640-pixel input (learning rate 10^{-4} , weight decay 10^{-4} , deterministic seed 20260807). The frozen condition schedule is 14 conditions per arm; the RetinaNet FP32 and legacy arms predate codec control and carry 13.

S5 Inference and evaluation contract

YOLO11 blocks: 640-pixel letterbox input, confidence threshold 0.001, class-wise NMS at IoU 0.7, at most 300 detections. RetinaNet blocks: 640-pixel letterbox input, confidence threshold 0.05, batched NMS at IoU 0.5, at most 300 detections. All evaluations use the COCO protocol via `pycocotools` with `maxDets=100`; VOC and KITTI annotations were converted once to COCO format and frozen. Decoder constants differ across families and are held fixed within each; cross-family comparisons are comparisons of recipe bundles, never of formats alone.

S6 Corruption-aware calibration schedule

The intervention calibration sets reuse the same 512 images as the matched clean-calibration manifests. Each image is deterministically assigned—via a SHA-256-keyed hash of its identifier—to exactly one of six in-family cells, Gaussian noise or JPEG at severities 1, 3, or 5 (approximately 77–102 images per cell). Assigned corruptions use the frozen transform parameters above, and the outputs are terminally encoded at JPEG quality 95, matching the evaluation input contract. In this first fold fog and motion blur never appear in calibration and form the held-out split. A complementary second fold (schedule `nn-corruptcalib-fold2-v1`, frozen before any of its runs) assigns the same images to fog or motion blur at severities 1, 3, or 5 (75–95 images per cell) and holds out Gaussian noise and JPEG. Its pre-specified rules are (i) a held-out gain over INT8-matched with an interval excluding zero and (ii) a positive held-out ΔE with an interval excluding zero, against both INT8-matched and INT8-q95calib.

The codec-matched control manifests encode the same 512 images at JPEG quality 95 with no named corruption, isolating codec adaptation from corruption coverage.

S7 Paired bootstrap

Within each dataset×model block, one deterministic schedule of $B = 2,000$ resamples of image indices is shared across all arms and all conditions, so every contrast is evaluated inside common-image samples. Reported point estimates are full-sample plug-in values; reported intervals are the 2.5th–97.5th percentiles of the bootstrap distribution of each contrast. Intervals cover image resampling only; they do not cover training-seed, calibration-draw, engine-build, or corruption-realization variance, and image-level resampling treats frames as independent (KITTI frames from shared driving sequences are correlated, so those intervals may be optimistic).

Two-sided bootstrap p -values are $p = \min\{1, 2 \min[\Pr(d \leq 0), \Pr(d \geq 0)]\}$ (resolution 1/2000) and are Holm-adjusted across the nine confirmatory YOLO11 blocks. Heterogeneity of ΔE across those blocks uses Cochran’s

$Q = \sum_i w_i (\hat{\Delta}_i - \bar{\Delta}_w)^2$ with $w_i = 1/\hat{\text{se}}_i^2$ from the bootstrap draws, $I^2 = \max\{0, (Q - \text{df})/Q\}$, and the DerSimonian–Laird between-block standard deviation τ . Blocks within a dataset share images, so their estimates are positively correlated; treating them as independent inflates the variance of between-block differences and makes Q and the pairwise z statistics conservative. RetinaNet and intervention intervals are per contrast.

Point estimates are full-sample plug-in values computed from the metric records of each cell (for COCO, from the frozen 2,000-image subset evaluation); bootstrap draws supply intervals, standard errors, and p -values only. For the intervention, the corruption-specific component on the held-out family is $\Delta E_{\text{held-out}} = (\text{AP}_A - \text{AP}_B)_{\text{held-out}} - (\text{AP}_A - \text{AP}_B)_{\text{J95}}$, the same paired construction as the main estimand. This decomposition is interpretive: it assumes an additive, parallel-shift model in which a calibration-induced level shift on clean inputs transfers comparably to corrupted inputs. Under an interaction model (where the intervention also changes slope across conditions), $\Delta E_{\text{held-out}}$ mixes that interaction with any clean-level effect; it is not a causal identification theorem. We therefore report the J95-clean and held-out levels alongside it rather than relying on the composite alone.

S8 Sensitivity analyses

Table S2 resolves the confirmatory ΔE of each YOLO11 block per corruption family (each entry is the family-mean FP8–INT8 gap minus the J95 gap) and gives the leave-one-out exclusion of the `motion_blur-s5` severity wall. The sign heterogeneity is also *within* blocks: on VOC-x the JPEG family contributes -1.07 AP while Gaussian noise and motion blur contribute $+0.7$ AP each; on COCO-m JPEG and motion blur contribute positively while noise contributes negatively. JPEG is therefore not a uniformly harmful or helpful family — it is the most block-variable family of the four. Excluding the severity wall leaves the cross-block heterogeneity essentially unchanged ($Q \approx 60.2$, $I^2 \approx 87\%$) and every Holm-surviving block’s sign intact; the two near-null KITTI blocks drift (YOLO11n flips from -0.01 to $+0.15$, still null, and YOLO11m’s negative effect loses significance). The interaction pattern is therefore not an artifact of the saturated cell, although the marginal KITTI estimates are sensitive to it.

Table S2: Per-family ΔE (AP points; family-mean FP8–INT8 gap minus J95 gap) and the leave-one-out ΔE excluding `motion_blur-s5`, per dataset×model block. * interval excludes zero.

Dataset	Model	Fog	Gaussian noise	JPEG	Motion blur	ΔE excl. mb-s5
KITTI	YOLO11n	+1.11*	+0.41	+0.17	−1.71*	+0.15 [−0.51, +1.03]
KITTI	YOLO11m	+0.38	−0.44	+0.02	−2.68*	−0.35 [−0.98, +0.29]
KITTI	YOLO11x	−0.40	−1.98*	−0.43	+0.03	−0.65*[−1.08, −0.29]
VOC	YOLO11n	+0.35*	+0.28	+0.27	+0.34*	+0.34*[+0.09, +0.62]
VOC	YOLO11m	−0.03	−0.48*	−1.23*	−0.56*	−0.53*[−0.76, −0.31]
VOC	YOLO11x	+0.26*	+0.69*	−1.07*	+0.67*	+0.13 [−0.06, +0.35]
COCO-2k	YOLO11n	+0.17	−0.22	−0.56*	−0.69*	−0.28 [−0.63, +0.05]
COCO-2k	YOLO11m	+0.36*	−0.36	+1.43*	+0.99*	+0.60*[+0.27, +0.97]
COCO-2k	YOLO11x	+0.15	−0.28	−2.10*	+0.13	−0.59*[−1.00, −0.26]

S9 Evidence layers

Diagnostic arms (the RetinaNet decomposition: head-only and all-except-region variants) are single-point recipe sweeps produced during development and carry no intervals. Confirmatory evidence is the frozen 14-condition paired protocol. Intervention evidence is the corrupt-calibration experiment together with its codec-matched controls. Historical exploratory results (a 144-cell grid over four datasets including TT100K, built under earlier engine contracts) are reported only as context and are never pooled with confirmatory estimates.

S10 Artifact and audit boundary

Manuscript tables and figures are generated deterministically from frozen ledgers by `analysis/build_nn_tables.py`. Engine registries, calibration manifests (including the corrupt-calibration cell-assignment schedule), input manifests, predictions, metric records, and paired-bootstrap draw caches are SHA-256 hash-bound and retained in the versioned research archive (concept DOI <https://doi.org/10.5281/zenodo.22031663>). Raw benchmark datasets, trained checkpoints, built TensorRT engines, and raw predictions are not redistributed; their provenance and regeneration path is documented in the archive.

S11 Minimum reporting checklist for paired corruption studies

- record the complete treatment recipe (checkpoint, quantizer coverage, calibration list and preprocessing, builder settings, decoder), not just the datatype label;
- pair treatments on identical image bytes, including terminal encoding;
- subtract the matched clean gap before reading a corruption interaction;
- report absolute corrupted AP and the clean gap alongside every interaction, so floor effects cannot masquerade as robustness;
- keep calibration corruption families disjoint from held-out evaluation families when claiming calibration-based robustness;
- carry codec-matched calibration controls before attributing calibration-intervention gains to corruption coverage;
- flag severity-wall cells where every treatment including FP32 is at the floor, and report a leave-one-out sensitivity.